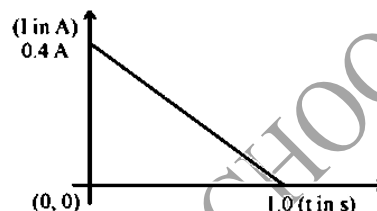


FREEDOM INTERNATIONAL SCHOOL
WORKSHEET
PHYSICS
CLASS XII
ELECTROMAGNETIC INDUCTION

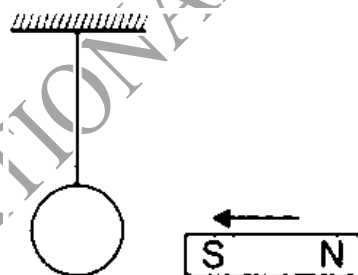
1. When a conducting loop of resistance $10\ \Omega$ and area $10\ \text{cm}^2$ is removed from an external magnetic field acting normally, the variation of induced current in the loop with time is shown in the figure.

Find the

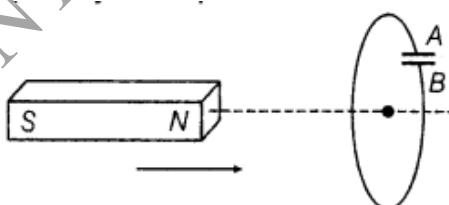
- (i) total charge passed through the loop.
- (ii) change in magnetic flux through the loop.
- (iii) magnitude of the magnetic field applied.



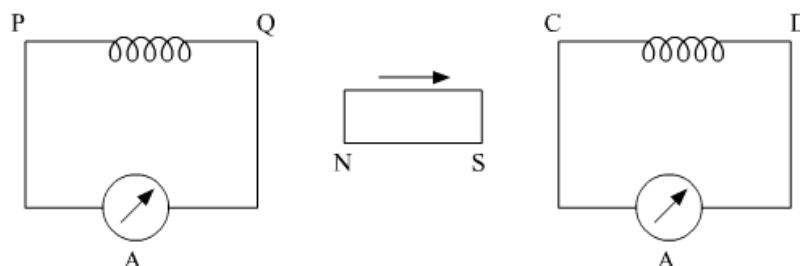
2. Give the direction in which the induced current flows in the wire loop, when the magnet moves towards it as shown in figure. How will the position of the loop be affected?



3. Predict the polarity of the plate A of the capacitor, when a magnet is moved towards the coil, as shown in the figure.

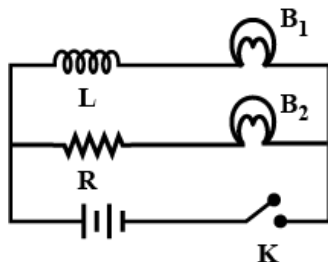


4. A bar magnet is moved in the direction indicated by an arrow between two coils PQ and CD as shown in the figure. Indicate the direction of the induced current in each coil.

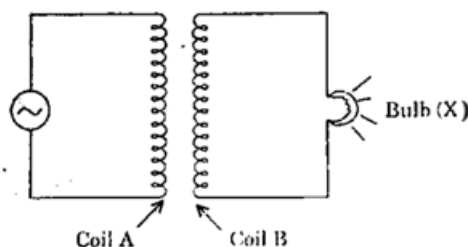


5. The figure shows an inductor L and a resistor R connected in parallel to a battery through a switch. The resistance of R is the same as that of the coil that makes L . Two identical bulbs are put in each arm of the circuit.
- (i) Which of the bulbs light up bright when K is closed?

- (ii) Will the two bulbs be equally bright after some time?
Give a reason for your answer.

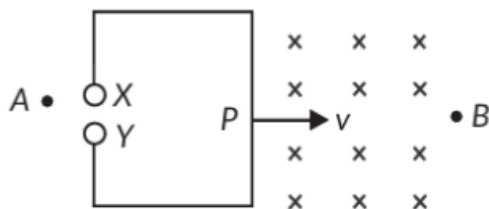


6. How does the self-inductance of an air core coil change, when (i) the number of turns in the coil is decreased, (ii) an iron rod is introduced in the coil?
7. Figure given below shows an arrangement by which current flows through the bulb (X) connected with coil B, when a.c. is passed through coil A.



Explain the following observations:

- (a) Bulb lights up.
(b) Bulb gets dimmer if the coil B is moved upwards.
8. A circular coil of radius 10 cm, 500 turns and resistance $2\ \Omega$ is placed with its plane perpendicular to the horizontal component of the earth's magnetic field. It is rotated about its vertical diameter through 180° in 0.25 s. Estimate the magnitudes of the emf and current induced in the coil. Horizontal component of the earth's magnetic field at the place is $3 \times 10^{-5}\text{ T}$.
9. If the current in the primary circuit of a pair of coils changes from 5 A to 1 A in 0.02 s, calculate (i) induced emf in the secondary coil if the mutual inductance between the two coils is 0.5 H and (ii) the change of flux per turn in the secondary, if it has 200 turns.
10. A rectangular coil P is moved from a point A to another point B with uniform velocity v through a region of a uniform magnetic field acting normally inwards as shown in the figure. Show graphically (i) the variation of magnetic flux associated with the coil with time, (ii) variation of induced emf across points X and Y of the coil with time.



Explain the nature of variation in magnetic flux as represented by the graph in first case.